

YIWEI YANG

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EDUCATION

UC Santa Cruz, Ph.D. Student 08/2022 – 06/2028

- Major: Computer Science, advised by Andi Quinn. 22 Fall TA of Computer Architecture

ShanghaiTech University, Undergraduate 09/2018 – 06/2022

- Major: Computer Science, finished Compiler, Network, Database, OS, CA, Convex, RL, Parallel Computing. 21& 22 Spring TA of Compiler

WORK EXPERIENCE

Oxmiq Labs, Campbell, CA 06/2025 – 09/2025

(Compiler Engineer) Compiler Engineer Intern

- Applied HetGPU into Production.

Jump Trading, Shanghai, China 07/2020 – 09/2020

(Linux Team) Production Engineer Intern

- High Frequency Trade Order Book simulation applying Linear Regression Method.
- Designed a user interface to automate the core affinity of jobs.
- Designed an eBPF exporter of GPFs full OSS lifetime traces for better reporting the bottleneck.

RESEARCH EXPERIENCE

Storage Systems Research Center, UC Santa Cruz 08/2022 – Present

(Graduate Research) Assistant

- Data-driven CXL.mem allocation, prefetching, and replacement policies Operating System. 🏆
- Make Hardware Software Co-design for on CXL.cache data movement 📦
- Make universal migration based on WebAssembly and PTX^{MV}_{VM}
- Make cross kernel-userspace eBPF observation for distributed system 🐝
- Contributed to vLLM/SGLang for high-throughput serving, implemented custom CUDA CXL kernels for attention mechanisms, optimized KV cache management and continuous batching strategies for improved GPU utilization. 🧵

SELECTED PUBLICATIONS

“GPUOS: A GPU Operating System Primitive for Transparent Operation Fusion” Preprint **Yiwei Yang**, Xiangyu Gao, Yuan Zhou, Yuhang Gan, Yusheng Zheng, Andi Quinn.

“DAXFS: A Lock-Free Shared Filesystem for CXL Disaggregated Memory” Preprint Cong Wang, **Yiwei Yang**, Yusheng Zheng.

“WIO: Upload-Enabled Computational Storage on CXL SSDs” Preprint **Yiwei Yang**, Yanpeng Hu, Yusheng Zheng, Estabon Ramos, Jianchang Su, Andi Quinn, Wei Zhang.

“gpu_ext: Extensible OS Policies for GPUs via eBPF” Preprint Yusheng Zheng, Tong Yu, **Yiwei Yang**, Minghui Jiang, Xiangyu Gao, Jianchang Su, Yanpeng Hu, Wenan Mao, Wei Zhang, Dan Williams, Andi Quinn.

“CXLaimPod: CXL Memory is all you need in AI era” Preprint **Yiwei Yang**, Yusheng Zheng, Yiqi Chen, Zheng Liang, Kexin Chu, Zhe Zhou, Andi Quinn, Wei Zhang.

“CXLMemUring: Revolutionizing Memory Access Through Compiler-Driven Heterogeneous Execution” Preprint **Yiwei Yang**, Yusheng Zheng, Yuyi Li, Jianchang Su, Wei Zhang, Andi Quinn.

“hetGPU: Pushing pooling into heterogeneous GPU Domain” HCDS26 **Yiwei Yang**, Yusheng Zheng, Tong Yu, Andi Quinn.

“CXLMemSim: Practical Performance Simulation and Characterization of CXL 3.0 Memory Systems” HPDC26 **Yiwei Yang**, Pooneh Safayanikoo, Jiacheng Ma, Tanvir Ahmed Khan, Andi Quinn.

“MVVM: Deploy Your AI Agents-Securely, Efficiently, Everywhere” Preprint **Yiwei Yang**, Aibo Hu, Yusheng Zheng, Brian Zhao, Xinqi Zhang, Andi Quinn.

“ChainIO: Bridging Disk and Network Domains with eBPF” eBPF25 Zheng Cao, Xuhang He, Yanpeng Hu, Yusheng Zheng, **Yiwei Yang**, Jianchang Su, Wei Zhang, Andi Quinn.

“Extending Applications Safely and Efficiently” OSDI25 Yusheng Zheng, Tong Yu, **Yiwei Yang**, Yanpeng Hu, Xiaozheng Lai, Dan Williams, Andi Quinn.

“wBPF: Efficient Edge-Case Observability for CXL Pooling systems via eBPF” HCDS25 Yusheng Zheng, Yu Tong, **Yiwei Yang**, Andi Quinn.

“Kgent: Kernel Extensions Large Language Model Agent” eBPF24 Yusheng Zheng, **Yiwei Yang**, Maolin Chen, Andi Quinn.

”Attack as Defense: Characterizing Adversarial Examples using Robustness” ISSTA21 Zhe Zhao, Guangke Chen, Jingyi Wang, **Yiwei Yang**, Fu Song, Jun Sun.

SKILLS

- **Programming Languages:** not limited to any specific language, and experienced in Python/C++/Rust, comfortable with Golang/C/Java/Scala/TypeScript (in random order).
- **System:** Specialist in Compiler & Performance Analysis, familiar with LLVM, MLIR, Gem5, WASM, gdb, eBPF, ptx, qemu, chisel, Linux mm & observability subsystem.
- **Machine Learning:** familiar with general knowledge of machine & reinforce learning, interested of System for LLM for Sys.

MISCELLANEOUS

- Interests: HPC, Architecture, Security, JIT, CXL, LLM4Sys4LLM, etc.
- Lead GeekPie_HPC *Ranked 2*, SC-SCC21. *Ranked 4*, ISC22. Advise Not-Slow-Slug *Ranked 2*, ISC23.
- Advise Kiki Zhang, Zhen Cao to be Ph.D. students.
- Google Summer of Code Mentor for BPFTime